HW5

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$$1.\ \Psi = A e^{i\omega(px+\eta_1^{\beta} z-t)} \qquad \vec{u}^{+} = \left[(\varphi_r)_x - \Psi_z - (\Psi_r)_z\right]\hat{x}$$

$$\Psi_r = A_r e^{i\omega(px-\eta_1^{\beta} z-t)} \qquad +\left[(\varphi_r)_z + \Psi_x + (\Psi_r)_x\right]\hat{z}$$

$$\Psi_t = A_t e^{i\omega(px+\eta_2^{\beta} z-t)} \qquad \vec{u}^{-} = \left[(\varphi_t)_x - (\Psi_t)_z\right]\hat{x}$$

$$\varphi_r = \cancel{A_r} B_r e^{i\omega(px-\eta_1^{\alpha} z-t)} \qquad +\left[(\varphi_t)_z + (\Psi_t)_x\right]\hat{z}.$$

$$\varphi_t = B_t e^{i\omega(px+\eta_2^{\alpha} z-t)}$$

边界条件: $\vec{u}^{+} = \vec{u}^{-}$, $\sigma_{31}^{+} = \sigma_{31}^{-}$. $\sigma_{33}^{+} = \sigma_{33}^{-}$.

$$\sigma_{31} = 2\mu e_{31} = \mu\left(\frac{\partial u_3}{\partial x} + \frac{\partial u_1}{\partial z}\right)$$

$$\sigma_{33} = \lambda(e_{33}+e_{11}) + 2\mu e_{33} = (\lambda+2\mu)\frac{\partial u_3}{\partial z} + \lambda\frac{\partial u_1}{\partial x}$$

$$\vec{u}^{+} = i\omega\left[B_r p e^{i\omega(px-\eta_1^{\alpha} z-t)} - A\eta_1^{\beta} e^{i\omega(px+\eta_1^{\beta} z-t)} + A_r\eta_1^{\beta} e^{i\omega(px-\eta_1^{\beta} z-t)}\right]\hat{x}$$

$$+ i\omega\left[-B_r\eta_1^{\alpha} e^{i\omega(px-\eta_1^{\alpha} z-t)} + A p e^{i\omega(px+\eta_1^{\beta} z-t)} + A_r p e^{i\omega(px-\eta_1^{\beta} z-t)}\right]\hat{z}$$

$$\vec{u}^{-} = i\omega\left[B_t p e^{i\omega(px+\eta_1^{\alpha} z-t)} - A_t\eta_2^{\beta} e^{i\omega(px+\eta_2^{\beta} z-t)}\right]\hat{x}$$

$$+ i\omega\left[B_t\eta_2^{\alpha} e^{i\omega(px+\eta_2^{\alpha} z-t)} + A_t p e^{i\omega(px+\eta_2^{\beta} z-t)}\right]\hat{z}$$

$$\cancel{\sigma_{31}^{+} = -\mu_1\omega^2\left[-B_r\eta_1^{\alpha} p e^{i\omega(px-\eta_1^{\alpha} z-t)} + A p^2 e^{i\omega(px+\eta_1^{\beta} z-t)} + A_r p^2 e^{i\omega(px-\eta_1^{\beta} z-t)}\right]}$$

$$\cancel{-B_r p\eta_1^{\alpha} e^{i\omega(px-\eta_1^{\alpha} z-t)}}$$

$$\sigma_{31}^{+} = \mu_1\left[(\varphi_r)_{zx} + \Psi_{xx} + (\Psi_r)_{xx} + (\varphi_r)_{zx} - \Psi_{zz} - (\Psi_r)_{zz}\right]$$

$$= \mu_1\left[2(\varphi_r)_{zx} + \Psi_{xx} + (\Psi_r)_{xx} - \Psi_{zz} - (\Psi_r)_{zz}\right]$$

$$= -\mu_1\omega^2\left[-2B_r p\eta_1^{\alpha} e^{i\omega(px-\eta_1^{\alpha} z-t)} + A(p^2-\eta_1^{\beta 2}) e^{i\omega(px+\eta_1^{\beta} z-t)} + A_r(p^2-\eta_1^{\beta 2}) e^{i\omega(px+\eta_1^{\beta} z-t}\right.$$

$$\sigma_{31}^{-} = \mu_2\left[2(\varphi_t)_{zx} + (\Psi_t)_{xx} - (\Psi_t)_{zz}\right]$$

$$= -\mu_2\omega^2\left[2B_t p\eta_2^{\alpha} e^{i\omega(px+\eta_2^{\alpha} z-t)} + A_t(p^2-\eta_2^{\beta 2}) e^{i\omega(px+\eta_2^{\beta} z-t)}\right]$$

$$\sigma_{33}^{+} = (\lambda + 2\mu)\frac{\partial u_3}{\partial z} + \lambda\frac{\partial u_1}{\partial x}$$

$$= (\lambda+2\mu)\left[(\varphi_r)_{zz} + \bar{\Psi}_{xz} + (\bar{\Psi}_r)_{xz}\right] + \lambda\left[(\varphi_r)_{xx} - \bar{\Psi}_{xz} - (\bar{\Psi}_r)_{xz}\right]$$

$$= (\lambda+2\mu)(\varphi_r)_{zz} + \lambda(\varphi_r)_{xx} + 2\mu\left(\bar{\Psi}_{xz} + (\bar{\Psi}_r)_{xz}\right).$$

$$\sigma_{33}^{-} = (\lambda+2\mu)\left[(\varphi_t)_{zz} + (\bar{\Psi}_t)_{xz}\right] + \lambda\left[(\varphi_t)_{xx} - (\bar{\Psi}_t)_{zx}\right]$$

$$= (\lambda+2\mu)(\varphi_t)_{zz} + \lambda(\varphi_t)_{xx} + 2\mu(\bar{\Psi}_t)_{xz}.$$

$$\sigma_{33}^{+} = \cancel{(\lambda_1+2\mu_1)\left[B_r\eta_1^{2\alpha}e^{(\ )} +\right.}$$

$$(\lambda_1+2\mu_1)B_r\eta_1^{2\alpha}e^{(\ )} + \lambda B_r p^2 e^{(\ )} + 2\mu\left[A p\eta_1^{\beta}e^{(\ )} \ast - A_r p\eta_1^{\beta}e^{(\ )}\right]$$

$$\sigma_{33}^{-} = (\lambda_2+2\mu_2)B_t\eta_2^{2\alpha}e^{(\ )} + \lambda B_t p^2 e^{(\ )} + 2\mu A_t p\eta_2^{\beta}e^{(\ )}$$

考虑位移势的透射反射比.

$$R_{ss} = \frac{A_r}{A}$$

$$R_{sp} = \frac{B_r}{A}$$

$$T_{ss} = \frac{A_t}{A}$$

$$T_{sp} = \frac{B_t}{A}$$

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$\cancel{B_r p e}$

$\cancel{B_r p e^{-i\omega\eta_1^\alpha z} - A\eta_1^\beta e^{i\omega\eta_1^\beta z} + A_r\eta_1^\beta e^{-i\omega\eta_1^\beta z} = B_t p e^{i\omega}}$

$z=0$ 处

$$B_r p - A\eta_1^\beta + A_r\eta_1^\beta = B_t p - A_t\eta_2^\beta$$

$$-B_r\eta_1^\alpha + Ap + A_r p = B_t\eta_2^\alpha + A_t p$$

$$\mu_1\left[-2B_r p\eta_1^\alpha + A(p^2-\eta_1^{\beta 2}) + A_r(p^2-\eta_1^{\beta 2})\right] = \mu_2\left[2B_t p\eta_2^\alpha + A_t(p^2-\eta_2^{\beta 2})\right]$$

$$(\lambda_1+2\mu_1)B_r\eta_1^{2\alpha} + \lambda B_r p^2 + 2\mu(A-A_r)p\eta_1^\beta = (\lambda_2+2\mu_2)B_t\eta_2^{2\alpha} + \lambda B_t p^2 + 2\mu A_t p\eta_2^\beta$$

$\Rightarrow$ ① $R_{sp}\,p \cancel{=\eta_r^\beta} + R_{ss}\eta_1^\beta \cancel{=} - T_{sp}\,p + T_{ss}\eta_2^\beta = \eta_1^\beta$

② $-\cancel{B} R_{sp}\eta_1^\alpha + R_{ss}\,p \cancel{+} \overline{T}_{sp}\eta_2^\alpha - T_{ss}\,p = -p$

③ $-2\mu_1 p\eta_1^\alpha R_{sp} + \mu_1(p^2-\eta_1^{\beta 2})R_{ss} - 2\mu_2 T_{sp}\,p\eta_2^\alpha - \mu_2 \overline{T}_{ss}(p^2-\eta_2^{\beta 2}) = -\mu_1(p^2-\eta_1^{\beta 2})$

$\cancel{(\lambda_1+2\mu_1)\eta_1^{2\alpha} R_{sp} +}$

④ $\left[(\lambda_1+2\mu_1)\eta_1^{2\alpha} + \lambda p^2\right] R_{sp} \cancel{+} - 2\mu p\eta_1^\beta R_{ss} - \left[(\lambda_2+2\mu_2)\eta_2^{2\alpha} + \lambda p^2\right] T_{sp}$
$- 2\mu p\eta_2^\beta T_{ss} = -2\mu p\eta_1^\beta$

其中. $\mu = \rho\beta^2$. $\lambda+2\mu = \rho\alpha^2$.

$\eta_\alpha^2 + p^2 = \alpha^2$. $\eta_\beta^2 + p^2 = \beta^2$.

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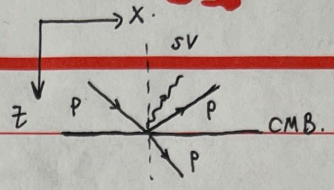


2. $\varphi = Be^{i\omega(px+\eta_1^\alpha z-t)}$

$\varphi_r = B_r e^{i\omega(px-\eta_1^\alpha z-t)}$

$\varphi_t = B_t e^{i\omega(px+\eta_2^\alpha z-t)}$.

$\Psi_r = A e^{i\omega(px-\eta_1^\beta z-t)}$.

$\vec{u}^+ = [\varphi_x + (\varphi_r)_x - (\Psi_r)_z]\hat{x} + [\varphi_z + (\varphi_r)_z + (\Psi_r)_x]\hat{z}$

$\vec{u}^- = (\varphi_t)_x \hat{x} + (\varphi_t)_z \hat{z}$.

边界条件. $u_3^+ = u_3^-$, $\sigma_{31}^+ = \sigma_{31}^-$. $\sigma_{33}^+ = \sigma_{33}^-$.

$\vec{u}^+ = i\omega[Bpe^{i\omega(px+\eta_1^\alpha z-t)} + B_r p e^{i\omega(px-\eta_1^\alpha z-t)} + A\eta_1^\beta e^{i\omega(px-\eta_1^\beta z-t)}]\hat{x}$

$+ i\omega[B\eta_1^\alpha e^{i\omega(px+\eta_1^\alpha z-t)} - B_r\eta_1^\alpha e^{i\omega(px-\eta_1^\alpha z-t)} + Ape^{i\omega(px-\eta_1^\beta z-t)}]\hat{z}$

$\vec{u}^- = i\omega B_t p e^{i\omega(px+\eta_2^\alpha z-t)}\hat{x} + i\omega B_t\eta_2^\alpha e^{i\omega(px+\eta_2^\alpha z-t)}\hat{z}$.

$\sigma_{31} = 2\mu e_{31} = \mu\left(\frac{\partial u_3}{\partial x} + \frac{\partial u_1}{\partial z}\right)$.

$\sigma_{33} = (\lambda+2\mu)\frac{\partial u_3}{\partial z} + \lambda\frac{\partial u_1}{\partial x}$.

①. $\cancel{Bp + B_r p + A\eta_1^\beta = B_t p}$. $B\eta_1^\alpha - B_r\eta_1^\alpha + Ap = B_t\eta_2^\alpha$.

①. $\cancel{B\eta_1^\alpha - B_r\eta_1^\alpha + Ap = B_t\eta_2^\alpha}$ (切向位移不连续. 即 $u_p^+ \neq u_p^-$).

② ③. $\mu_1(B\eta_1^\alpha p - B_r\eta_1^\alpha p + Ap^2 + Bp\eta_1^\alpha - B_r p\eta_1^\alpha - A\eta_1^\beta)$

$= \mu_2(B_t\eta_2^\alpha p + B_t p\eta_2^\alpha)$.

即 $\mu_1[2B\eta_1^\alpha p - 2B_r p\eta_1^\alpha + A(p^2 - \eta_1^{\beta 2})] = 2\mu_2 B_t\eta_2^\alpha p$.

③ ④. $(\lambda_1+2\mu_1)(B\eta_1^\alpha + B_r\eta_1^\alpha - Ap\eta_1^\beta) + \lambda_1(Bp^2 + B_r p^2 + A\eta_1^\beta p)$

$= (\lambda_2+2\mu_2)(B_t\eta_2^\alpha) + \lambda_2 B_t p^2$.

$$-B_r\eta_1^{\alpha}-B_t\eta_1^{\alpha}+Ap=-B\eta_1^{\alpha}.$$

$$-2p\eta_1^{\alpha}\mu_1B_r-2p\eta_2^{\alpha}\mu_2B_t+A(p^2-\eta_1^{2\beta})\mu_1=-2\mu_1p\eta_1^{\alpha}B$$

$$\left[(\lambda_1+2\mu_1)\eta_1^{2\alpha}-\lambda p^2\right]B_r-\left[(\lambda_2+2\mu_2)\eta_2^{2\alpha}+\lambda_2p^2\right]B_t+A\left[-(\lambda_1+2\mu_1)p\eta_1^{\beta}+\lambda_1\eta_1^{\beta}p\right]$$

$$=-\left[(\lambda_1+2\mu_1)\eta_1^{2\alpha}+\lambda_1p^2\right]B.$$

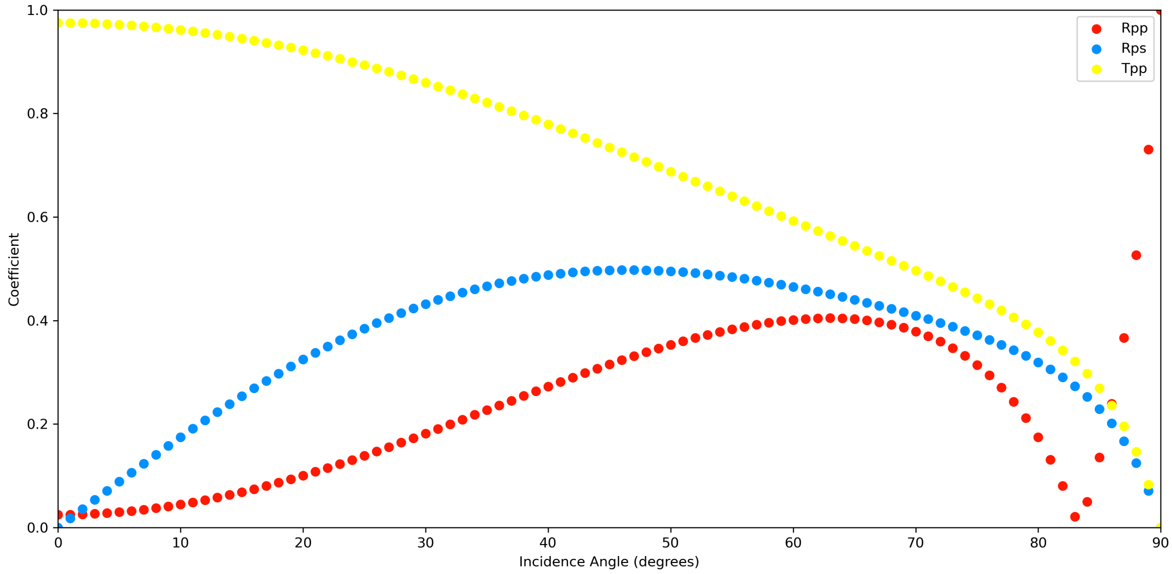
其中、位移与势函数的关系为 $|U_\alpha|=|k_\alpha|\varphi=\frac{\omega}{\alpha}\varphi$、$|U_\beta|=|k_\beta|\Psi=\frac{\omega}{\beta}\Psi$.

位移的透射反射系数：

$$R_{pp}=\frac{B_r}{B}.\qquad R_{ps}=\frac{\alpha_1\cancel{B_t}A}{\beta_1B}\qquad T_{pp}=\frac{\cancel{\alpha_1}B_t}{\cancel{\beta_2}B}\quad\frac{\alpha_1B_t}{\alpha_2B}$$

~~编写~~

编程求解方程，绘图得：



注意到，当入射角约83°时，P波反射后完全转换为SV波。

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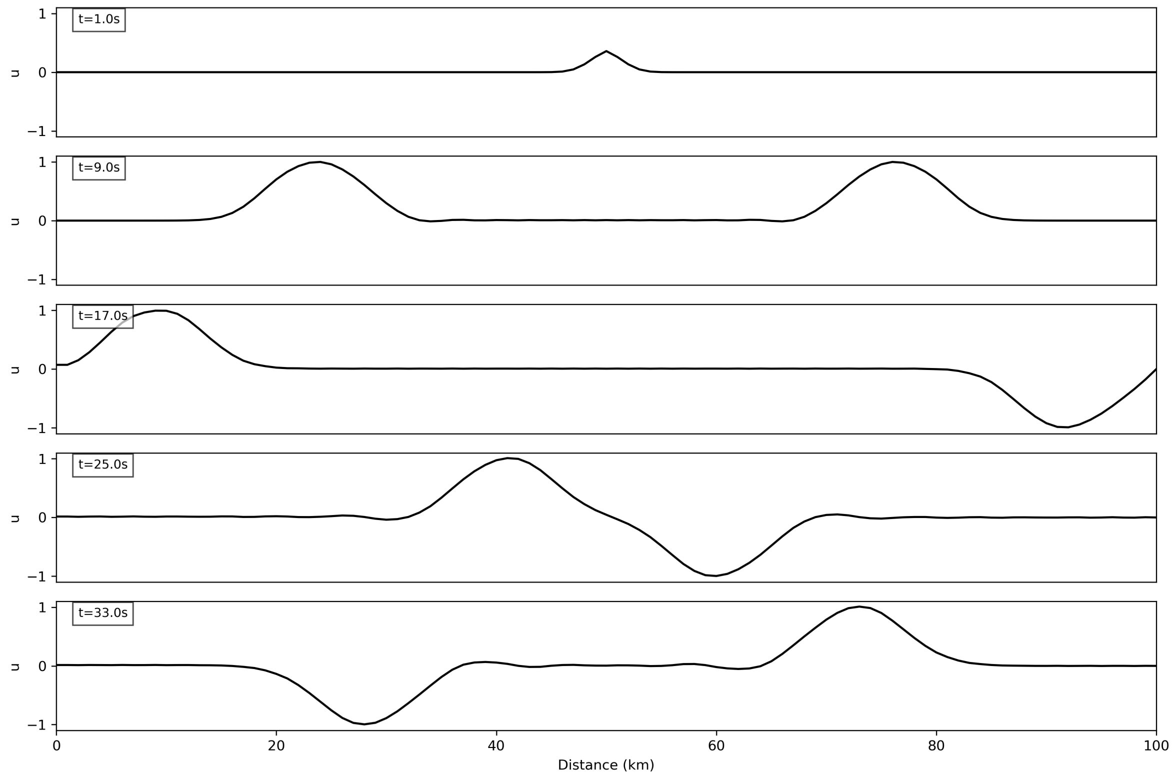
绘制了切片和GIF动图（后者见附件）。



由切片可见，

- 波的传播速度为4km/s。
- 在自由边界，波反射后不存在半波损失；在固定边界，波反射后存在半波损失。
- 波相遇时，存在波的干涉；波相遇离开后，波形不变。

由GIF动图可见，波存在明显的拖尾，且拖尾现象随着时间的推移越来越严重。拖尾是有限差分造成的数值色散现象，即不同频率的波在数值上的相速度不同。数值色散是二阶中心差分的固有缺陷，此题要求的差分网格较为稀疏，所以这一现象较为显著。